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THE PALAEOPOSITION OF MADAGASCAR: REMANENCE AND MAGNETIC PROPERTIES OF LATE PALAEOZOIC SEDIMENTS

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Palaeomagnetic results are reported from the predominantly green sediments of the Upper Permian to Lower Triassic Sakamena Group and the Upper Carboniferous to Lower Permian Sakoa Group of Madagascar. Secondary magnetizations could only be removed successfully through thermal demagnetization procedures and then only if the cleaning process was completed by 450°C. Heating in air caused extensive magnetochemical changes to occur above this temperature. Coercivity spectrum analysis and low-temperature characteristics of the heated and unheated green sediments indicate that considerable amounts of fine-grained single-domain magnetite are formed at 500°C or more from some non-magnetic mineral, probably the iron silicates. For this reason consistent palaeomagnetic data could only be obtained from about half the samples collected. Results from 4 sites (19 samples) of the Lower Sakamena Group give a palaeomagnetic pole at 64.9S, 113.9E ($A_{95} = 5.6°$) and 3 sites (16 samples) from the Glacial Series of the Sakoa Group give a pole at 47.9S, 84.1E ($A_{95} = 8.1°$). When compared with corresponding data from Africa these results confirm and strengthen our previous conclusions from the Triassic-Jurassic Isalo Group regarding the palaeoposition of Madagascar. All three poles are only consistent with the Smith and Hallam reconstruction which places Madagascar off the eastern coast of Africa adjacent to Kenya and Tanzania.

1. Introduction and sampling

In a previous paper [1] we presented some palaeomagnetic results from the Triassic-Jurassic Isalo Group, the youngest group comprising the Malagasy Karroo Supergroup. In this paper we present results from the underlying Sakoa and Sakamena Groups; they reinforce our previous conclusions regarding the palaeoposition of Madagascar adjacent to Africa in the supercontinent of Gondwanaland.

The geology of Madagascar has recently been reviewed by Besairie [2] who has also provided a more detailed description of the sedimentary sequences of the island [3]. The Karroo Supergroup forms a continuous north–south sedimentary belt on the western side, transgressively overlying the Precambrian crystalline basement eastwards (see inset of Fig. 1). The basal Sakoa Group of Late Carboniferous to Early Permian age is only developed in the southwest of the island. The lowest beds constitute the Glacial Series containing tillites which form part of the evidence for the great late Palaeozoic glaciation of Gondwanaland [4]. The Sakoa Group is overlain with an angular unconformity of about 10° by the Sakamena Group of Late Permian to Early Triassic age. The facies of the Sakoa and Sakamena Groups in the southwest are essentially continental, but contain important marine incursions. In the extreme north of the island, however, the Sakamena Group is entirely marine. A summary of the Sakoa and Sakamena sequences in southwest Madagascar is given in Table 1.

Previously Nairn [5] reported measurements of the natural remanent magnetization (NRM) of six samples of the Lower Red Series of the Sakoa Group. During September 1973 we collected 91 samples from both the Sakoa (57 samples) and Sakamena (34 samples) Groups in the southwest of the island. Unfortunately the 17 samples collected from the Lower Red Series of the Sakoa Group were mislaid in transit so that this report deals only with the remaining 74 samples

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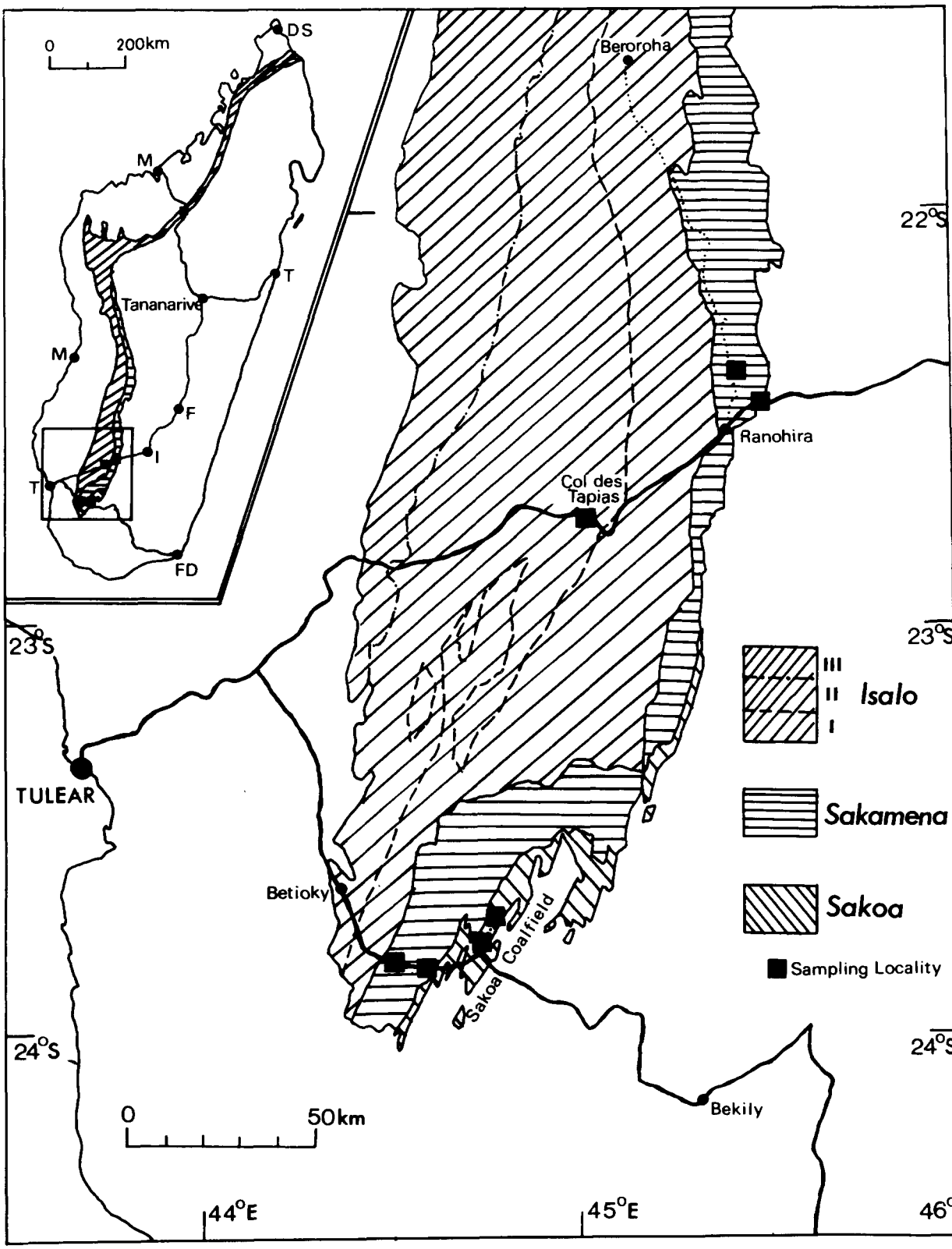


Fig. 1. The Karroo Supergroup in the southwest of Madagascar showing the sampling sites in the Sakoa and Sakamena Groups. Sampling sites from our previous study of the Isalo Group [1] are shown at the Col des Tapias and in the northwest of the island (inset).

TABLE 1

The Sakoa and Sakamena sequences of southwest Madagascar

Geological period	Horizons (thickness)		
Lower Triassic	Sakamena	6. Upper Red Series	–Upper (500 m)
		5. fish and ammonite bed	–Middle (200 m)
		4. shales, sandstones with reptiles	
Upper Permian		3. plant-bearing shales	
		2. lower marine bed	–Lower (2500 m)
		1. basal conglomerate	
	Angular unconformity		
Lower Permian	Sakoa	4. Vohitolia Limestones (10–20 m)	
		3. Lower Red Series (1400 m)	
		2. Coal Measures (100 m)	
Upper Carboniferous		1. Glacial Series (500 m)	

of our collection. Previously we have reported measurements on 32 samples from the Isalo Group [1].

Samples were collected from two regions (Fig. 1). Sections of the Lower Sakamena were sampled 11 km east of Ranohira on the main road from Tananarive to Tulear and also 11 km north of Ranohira on the track to Beroroha. The basal sediments immediately overlying the crystalline basement sampled on the main road are green sandstones and shales, but further north the higher horizons sampled also contain buff-coloured sandstones. The sediments in this region are almost flat lying with variable dips of only a few degrees. The Lower Sakamena and the Glacial Series of the Sakoa Group were sampled in the region of the Sakoa Coalfield east of Betioky (Fig. 1). Here the sediments dip towards the northwest with dips varying between 15 and 30°. The Lower Sakamena here consists mainly of green sandstones and shales, but samples collected from the Glacial Series of the Sakoa Group included also some dark shales near the base interbedded with the tillites. Towards the top of the Glacial Series a 20-m section immediately underlying the Coal Measures was sampled with 8 samples varying in lithology from green to black shales.

2. Palaeomagnetic measurements

The directions and intensities of remanent magnetization were measured using a DIGICO Balanced Fluxgate Rock Magnetometer [6]. Four specimens cut from each oriented sample were processed. The NRM directions almost all clustered around the direction of the present earth's field. Intensities of magnetization were generally around 1 mA m^{-1} (1 X 10^{-6} emu cm^{-3}) but a number of samples had initial intensities much less than this leaving little prospects for successful demagnetization experiments. Thermal demagnetization by the progressive method was carried out in a magnetic field-free space automatically maintained to within 1 nT (1 gamma) [7]. All directions of magnetization were analysed using the statistical method of Fisher [8].

2.1. Sakamena Group

Pilot specimens responded to progressive thermal demagnetization in two ways. Some specimens very quickly moved to a downward-seeking direction in the southeast quadrant or an upward-seeking direction in the northwest quadrant and became stable at temperatures between 200 and 450°C. The other group of specimens either remained directed in the present geomagnetic field direction or else attempted to move towards a stable group. However, at temperatures of 500°C and higher the samples became unstable and no systematic behaviour was subsequently observed. This was also a feature of the first group of stable samples when heated to 500°C or more. Directions of magnetizations started to scatter and became randomly directed. The thermal demagnetization curves of the two groups of specimens are quite distinctive (Fig. 2). Specimens which reached stable directions had curves which flatten out between 100 and 400°C (Fig. 2a) whereas

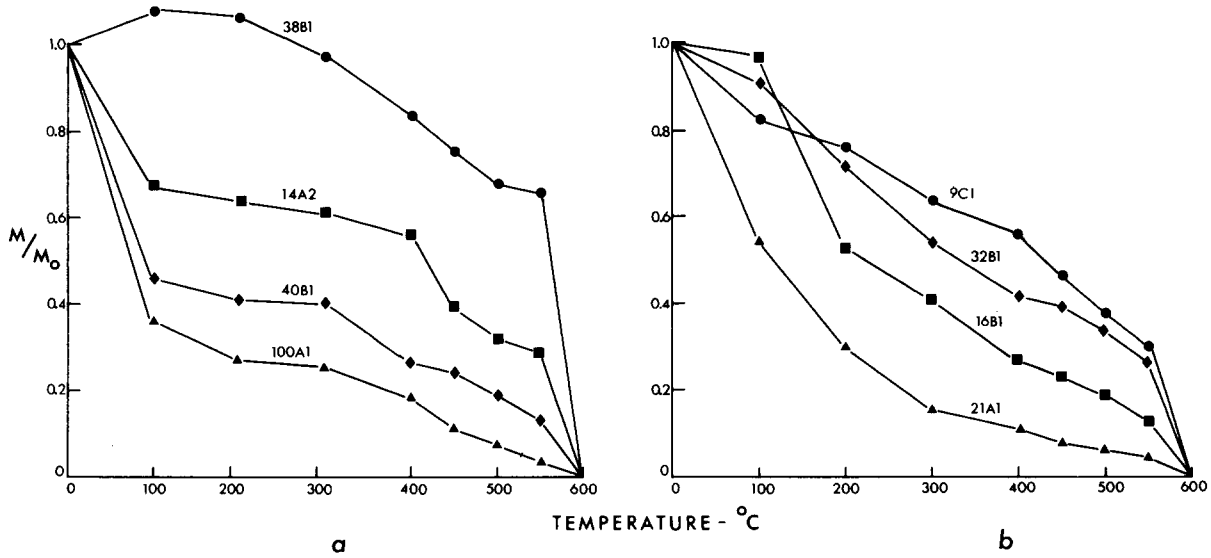


Fig. 2. Thermal demagnetization curves of samples from the Sakamena Group. (a) Specimens that were cleaned successfully. (b) Specimens from which no consistent data could be obtained.

those which failed to reach the stable directions had curves showing a gradual decrease to low intensities at 550 and 600°C (Fig. 2b).

The overall behaviour of the specimens appears to be related to some chemical change taking place at temperatures in excess of 450°C. The pilot specimens showed that no stable directions could be observed above this temperature. Since specimens from the same sample did not necessarily clean at the same temperature all specimens (4 from each sample) were successively heated to four temperatures – 200, 300, 400 and 450°C. The combination of four specimen directions within that temperature range that gave the highest value of Fisher's precision parameter k was regarded as the best estimate of the cleaned direction of magnetization for each sample. Only 19 of the 34 samples gave consistently grouped directions after this procedure (Fig. 3). Attempts were made to see if results could be obtained from the other 15 samples using alternating field demagnetization on further specimens, but these were unsuccessful.

The 19 samples have been divided into four sites as shown in the analysis of Table 2. Sites 1 and 2 represent the two localities sampled near Ranohira. Site 1 which is reversely magnetized immediately overlies the crystalline basement and is therefore stratigraphically lower than site 2. In the Sakoa Coalfield, however, the normally magnetized samples are all lower in the sequence than the reversely magnetized ones. Sites 3 and 4 from this region are a subdivision of the samples ac-

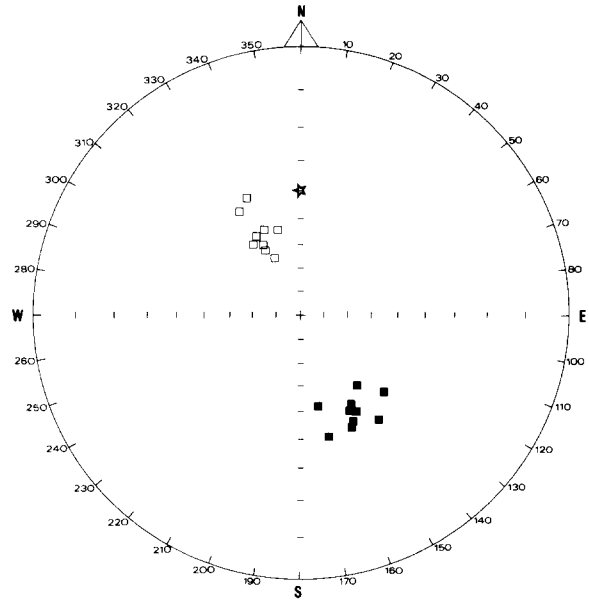


Fig. 3. Stereographic projection of directions of magnetization obtained from the 19 samples that could be cleaned from the Sakamena Group (squares). Solid symbols are on the lower hemisphere, open symbols on the upper hemisphere. The star is the present axial dipole field direction.

TABLE 2

Summary of palaeomagnetic data from the Lower Sakamena Group and the Glacial Series of the Sakoa Group after thermal cleaning

Site	Location		N	D, I	R	k	α_{95}	Pole position
(a) *Sakamena*								
1	22.5S	45.5E	7	151.6, +45.7	6.8903	54.7	8.2	63.8N, 299.4E
2	22.4S	45.4E	5	337.0, −55.3	4.9799	199	5.4	65.9N, 276.3E
3	23.8S	44.6E	4	331.9, −47.7	3.9572	70.1	11.0	64.3N, 297.2E
4	23.8S	44.5E	3	151.9, +45.3	2.9973	740	4.5	64.4N, 301.7E
	Mean of samples		19	153.0, +48.6	18.7707	78.5	3.8	–
	Mean of sites		4	152.9, +48.5	3.9889	270	5.6	–
	Mean of site poles		4	–	3.9886	263	5.6	64.9N, 293.9E
(b) *Sakoa*								
1	23.8S	44.7E	5	138.2, +65.5	4.8793	33.1	13.4	49.2N, 268.1E
2	23.8S	44.7E	3	132.1, +66.5	2.9446	36.1	20.8	44.9N, 268.2E
3	23.8S	44.7E	8	144.3, +70.5	7.8611	50.3	7.8	49.3N, 255.8E
	Mean of samples		16	139.7, +68.3	15.6650	44.7	5.5	–
	Mean of sites		3	137.8, +67.5	2.9963	539	5.3	–
	Mean of site poles		3	–	2.9912	228	8.1	47.9N, 264.1E

cording to polarity. Sample-mean and site-mean directions of magnetization were subjected to a two-tier analysis [9] to determine the significance of the between-site variation as judged by an F-test. In this case

$$F_{6,30} = \frac{\text{mean square between-sites}}{\text{mean square within-sites}} = 1.54$$

However, the upper 5% point ($P = 0.05$) for $F_{6,30} =$ = 2.42 so that the between-site variation is not significant. This indicates that the sample means are providing the same time-average as site means, a situation that we also observed for the Isalo Group [1]. There would be justification therefore in using the result obtained by averaging all 19 samples. In Table 2 correction has been made for the attitude of the beds at each site. Although this correction results in increased precision, it is not statistically significant.

2.2. *Sakoa Group*

The sediments sampled from the Glacial Series of the Sakoa Group were of much more variable lithology than those of the Lower Sakamena. Eight samples collected from the shales at the top of the Glacial Series showed behaviour on thermal demagnetization very similar to that of the stable samples from the Sakamena Group. All eight samples cleaned to consistent steeply inclined directions downwards towards the southeast in temperatures of 300 and 400°C (site 3 in Table 2). The remaining 32 samples from the lower sections of the Glacial Series exhibited erratic behaviour as was generally found for the unstable samples from the Sakamena Group. As with the Sakamena Group erratic directional changes always took place at 500°C or higher. A similar procedure was adopted as for the Sakamena samples and four specimens from each sample were heated to four successive temperatures between 200° and 450°C. Consistent directions from all four specimens could only be observed for eight of the remaining samples. The thermal demagnetization curves show similar characteristics (Fig. 4) to those of the Sakamena Group. Unstable samples showed no plateau in their curves and in all cases it appeared that some chemical change was occurring at 500°C or higher. Attempts at alternating field demagnetization again produced no consistent results.

The directions of magnetization of the 16 samples that could be cleaned are steeply dipping (inclination +68°) towards the southeast. All are reversely magnetized as illustrated in Fig. 5. The samples have been grouped into three sites (Table 2). Site 1 includes five

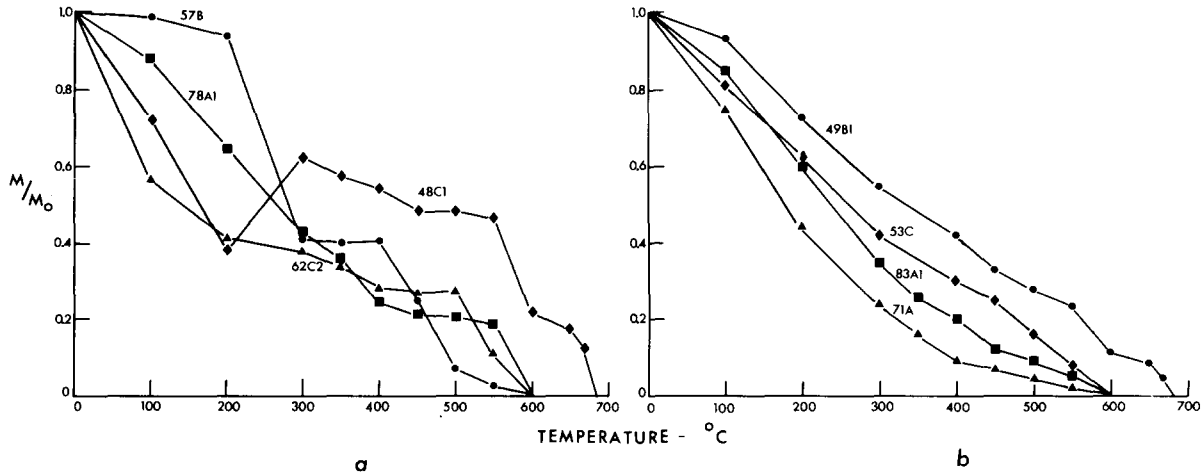


Fig. 4. Thermal demagnetization curves of samples from the Sakoa Group. (a) Specimens that were cleaned successfully (b). Specimens from which no consistent data could be obtained.

samples from the lower part of the same section as site 3 (8 samples) and site 2 contains three samples from a locality several kilometres further south. Application of an F-ratio test again shows that the between-site variation is not significant. After correcting for the attitude of the beds, the small increase in precision observed is not statistically significant.

The reversed polarity observed in all samples that gave results from the Sakoa Group is consistent with the predominantly reversed interval of about 50 m.y. that occurred during the late Palaeozoic [10]. The mean inclination of 68° suggests a palaeolatitude of 51° assuming an axial geocentric dipole field. The glacial environment suggested by the tillites of the Glacial Series must have existed during times when the polar ice cap was at least as extensive as that observed during the Pleistocene glaciation in the northern hemisphere.

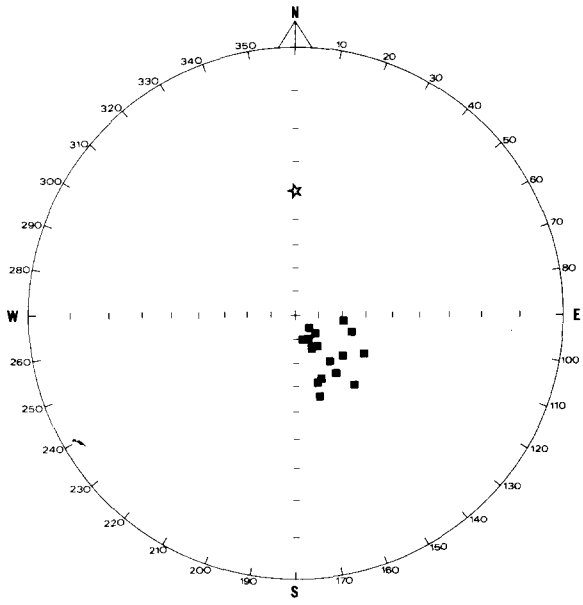


Fig. 5. Stereographic projection of directions of magnetization obtained from the 16 samples that could be cleaned from the Sakoa Group. Symbols are as in Fig. 3.

3. Magnetic properties of heated and unheated green sediments

The palaeomagnetism of continental sediments has most commonly involved studies of redbeds. For these rocks cleaning procedures make use of thermal demagnetization techniques as have been used in this study. The effect of such heat treatment on the magnetic mineralogy of red sediments has been investigated using coercivity spectrum analyses [11] or rotational hysteresis techniques [12]. Since it appears that the predominantly green sediments of the Sakoa and Sakamena Groups have undergone considerable magnetochemical changes when heated to temperatures above 450°C, we have carried out some experiments to investigate the effect of heat treatment on them.

A previous study of parallel sequences of red and green sediments has shown that each provided essentially the same record of geomagnetic field variations at the time of their deposition [13]. This was an important finding because it indicated that very different environments do not necessarily affect the ability of sediments to preserve the direction of the ancient geomagnetic field. Red sediments presumably result from formation in an oxidizing environment whereas green sediments result from formation in a more reducing environment.

In red sediments haematite is generally the main magnetic mineral although accessory magnetite is often present. The haematite occurs in two major forms, either as particles of black crystalline specularite (>1 μm) or as fine particles (<1 μm) of red pigment coating and cementing the other minerals. Collinson [27] has recently concluded that specularite is the most important carrier of remanent magnetization in red sandstones, contributing a very stable magnetization of depositional origin. Dunlop [11] proposes that the relative contributions of magnetite (or maghaemite), specular haematite and red pigment to IRM characteristics are indicated in an approximate way respectively by the proportions of soft (0–0.1 T), intermediate (0.1–0.3 T) and hard (>0.3 T), fractions in the coercivity spectrum (1 T = 10^4 gauss). In green sediments clearly the red pigment is absent and therefore the high-coercivity (>0.3 T) remanence fractions generally identified with haematite pigment can also be expected to be absent. Furthermore the reducing environment associated with green sediments would suggest magnetite rather than haematite to be the main magnetic mineral present in these sediments.

3.1. Coercivity spectrum analysis

It has been shown that the remanent coercivity spectrum can be derived from the isothermal remanence versus applied field characteristics (M_{ir}–B curves) of red sediments [11]. The coercivity spectrum can then be approximated by the incremental isothermal remanence ΔM in suitable intervals ΔB (0.1 T used by Dunlop [11]) as determined from the M_{ir}–B curve at room temperature. We have applied this technique to five representative specimens from each of the Sakamena and Sakoa Group sediments. As was expected virtually all the specimens had acquired their saturation isothermal remanence at a field of 0.5 T so that the high-coercivity fraction generally associated with the red pigment form of haematite was absent. For this reason the M_{ir}–B curves were only taken as far as 0.5 T for convenience. Specimens were first magnetized in increments of 0.05 T up to 0.5 T and were then heated to 700°C in air, cooled in field-free space and the measurements repeated.

Some dramatic changes in M_{ir} at 0.5 T were noted after the specimens were heated with increases ranging up to 770 times the unheated value! The greatest variation occurred within the Sakoa specimens which probably explains the greater difficulty encountered in isolating the primary remanence and the more varied forms of thermal demagnetization curves obtained (Fig. 4). Fig. 6 shows two M_{ir}–B curves obtained from specimens of the Sakoa Group. Specimen 62A2 is from a green shale occurring near the top of the Glacial Series. After heating, its isothermal remanence at 0.5 T increased by a factor of 249 from 88 mA m^{-1} to 21,900 mA m^{-1}. The coercivity spectrum (in 0.05-T increments) shows an enhancement at the lower end (where the 0–0.05 T range has increased by a factor of 312) relative to the higher part above 0.1 T. At the same time the bulk coercivity of the 0.5-T remanence has decreased from 60 to 45 mT. The unheated coercivity spectrum shows clearly that the major magnetic mineral is magnetite and that specularite can only be a minor accessory mineral. The very large increase in remanence mainly at the low-coercivity end of the spectrum must therefore be due to the formation of large quantities of magnetite. This can only have been derived from some originally non-magnetic mineral, a conclusion similar to that found by Dunlop [11] when red sediments are heated above 500°C. Our observations on green sediments concur with his findings for red sediments but contrast with the experiments of Schwarz [14] in which no magnetite production was found to have taken place in the green inclusions in some red sediments after heating.

The magnetic characteristics outlined for specimen 62A2 were observed in all the green sediments investigated. Heating to 700°C in air caused increases in the saturation remanence (0.5 T) of from twice to several hundred times the unheated value. Susceptibility increases varied from twice to 30 times and there was always an increase in the low-coercivity fraction (0–0.05 T) by factors that correlated with the increase in satura-

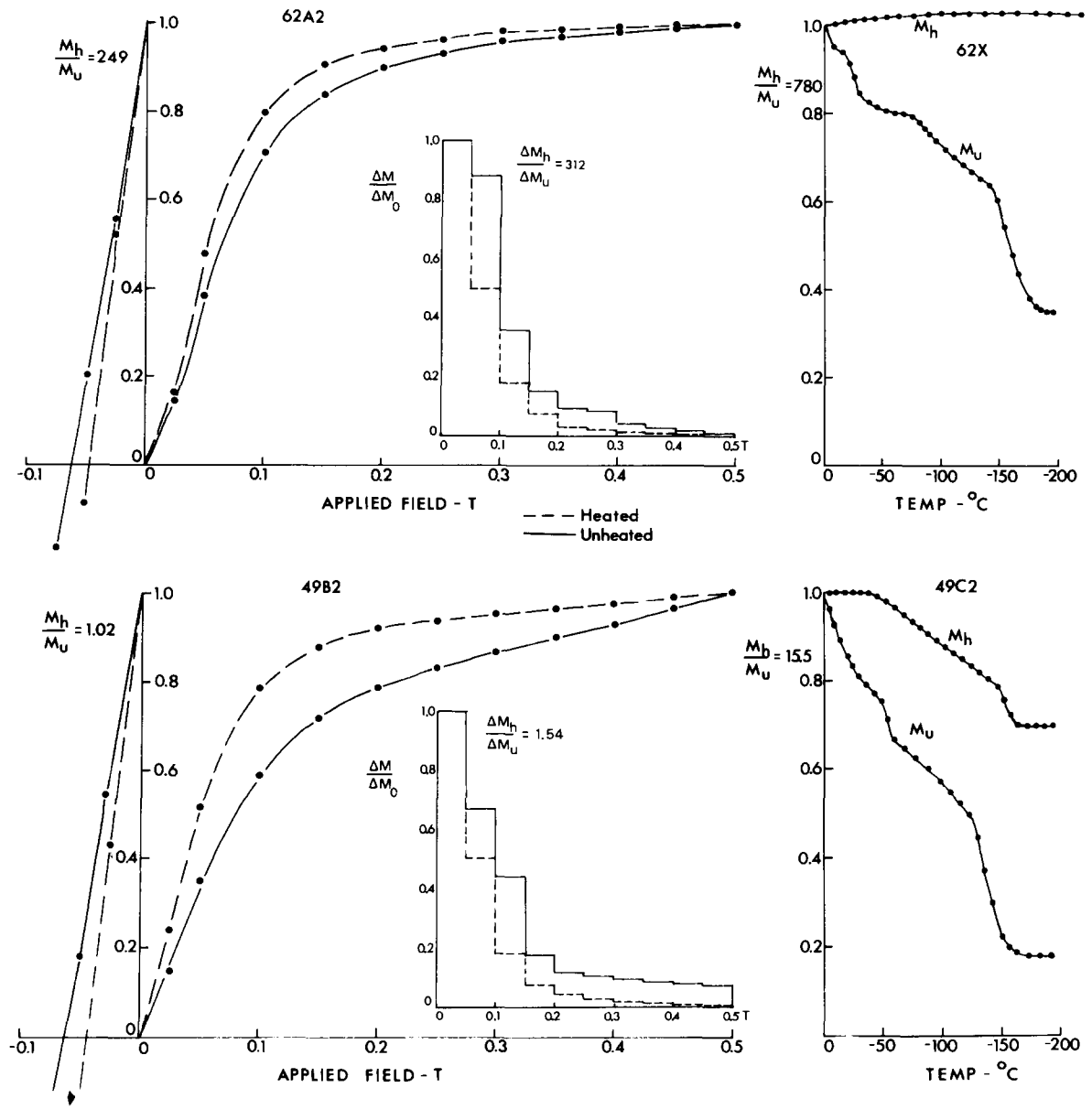


Fig. 6. Comparison of isothermal remanent magnetization curves of specimens from the Sakoa Group in their original state (solid lines) and after being heated to 700°C (dashed lines). Sample 62 is a green shale from the upper part of the Glacial Series and sample 49 is from near the base. The curves are normalized to the unheated (M_u) and heated (M_h) remanences at 0.5 T. The coercivity spectrum, derived from the incremental isothermal remanence ΔM in intervals of 0.05 T, is shown as the inset on the left-hand side. Both the unheated and heated spectra are normalized to the contribution in the lowest interval (0–0.05 T). On the right-hand side the variation of remanent magnetization on cooling to liquid nitrogen temperature (–196°C) is compared for unheated (NRM) and heated (TRM) specimens from the same samples.

tion remanence. A corresponding decrease in bulk coercivity of saturation remanence was also observed. It is clear that the failure to obtain any sensible directional information from these green sediments after heating to temperatures in excess of 450°C relates to the observed magnetochemical changes accompanied

by the production of large quantities of magnetite.

Specimen 49B2 is a dark-grey shale from the lower part of the Sakoa Group and comes from one of the few samples that showed any remanence remaining on thermal demagnetization above the Curie point of magnetite (Fig. 4). Although not typical of the magnetic characteristics of the collection as a whole its magnetic properties are of some interest. The presence of haematite is confirmed by the obvious existence of coercivities in excess of 0.5 T (Fig. 6). After heating there is only a 2% increase in the 0.5-T isothermal remanence but the 0–0.05 T coercivity range has been enhanced by more than 50%. This is accompanied by a marked decrease in the coercivity spectrum above 0.2 T, and a decrease in the bulk coercivity of the 0.5-T remanence from 62.5 to 45 mT. Again there must have been some magnetite production from non-magnetic minerals, but the change in the higher parts of the coercivity spectrum suggests at least some of the magnetite may have been derived from the reduction of haematite. However, grain growth in fine particle haematite can also lower its coercivity [15], a point which will be investigated further from the low-temperature experiments described below.

3.2. Low-temperature characteristics

The use of low-temperature transitions has been proposed for the identification of magnetite and haematite in rocks [16]. Magnetocrystalline anisotropy transitions occur at temperatures across which there are changes in the direction of the easy axes of magnetization. In magnetite the easy axes change from body diagonals (high temperature) to cube axes (low temperature) near −140°C. In haematite the easy axis changes from the basal plane (high temperature) to the ternary axis (low temperature) near −20°C, known as the Morin transition. However, these transitions are suppressed in fine particles of both haematite and magnetite. In magnetite this is because only multidomain grains are affected by low-temperature treatments since shape anisotropy dominate single-domain magnetite [17].

The NRMs of a further ten test specimens from the samples being investigated were continuously measured down to liquid nitrogen temperatures (−196°C) using an attachment to the Digico Magnetometer. The specimens were then heated in air to 700°C and cooled in the earth's magnetic field and their low-temperature characteristics measured. None of the Sakamena Group specimens showed any low-temperature transitions either before or after heating. Presumably this indicates predominantly fine-grained magnetite was originally present and was also produced during the magnetochemical changes taking place on heating. Specimens 62X and 49C2 from the Sakoa Group showed the cooling curves shown in Fig. 6.

The unheated green shale (62X) shows a small Morin transition around −25°C and a marked magnetite transition at around −150°C. After heating no low-temperature transitions are observed at all. Therefore although the coercivity spectrum analysis of its sister specimen 62A2 indicates that very large amounts of magnetite have been produced on heating, the low-temperature characteristics of these new particles swamps those of the original particles because of the two-hundred-fold increase in saturation isothermal remanence. The unheated dark-grey shale (49C2) also shows a Morin transition and a magnetite transition around −140°C. After heating, the Morin transition has disappeared and the magnetite transition, although present, is less pronounced. The sister specimen 49B2 showed virtually no change in the 0.5-T isothermal remanence after heating but a marked change in the coercivity spectrum that suggested magnetite had been produced from the reduction of haematite. The disappearance of the Morin transition after heating of specimen 49C2 confirms that this must be the case.

3.3. Discussion

Fine-grained magnetite is the main contributor to the remanence in the predominantly green sediments of the Sakoa and Sakamena Groups. Minor amounts of haematite may be present especially in some of the dark-grey shales of the Sakoa Group. Upon heating in air to 700°C there is considerable production of magnetite from originally non-magnetic minerals, probably iron silicates, sometimes resulting in increases of two orders of magnitude in saturation isothermal remanence. The magnetite is also fine-grained and single-domained. In some of the dark-grey shales reduction of haematite to magnetite is also observed. These magnetochemical changes taking place during thermal demagnetization do not necessarily mean that no results can be obtained. The difficulty is when the partial

thermoremanence (PTRM) acquired by any new material dominates the NRM of the rock or the material carrying the NRM is altered or destroyed. In the present study a dominant PTRM in the new magnetite is presumably the trouble. For this reason it was never possible to obtain results from thermal cleaning if temperatures in excess of 450° were required for the procedure. Therefore only those samples whose secondary magnetization could be removed by 450° were able to produce viable palaeomagnetic results. Above this temperature the onset of secondary magnetite production occurs rendering the sample useless for obtaining palaeomagnetic information. This explains why only a fraction (56% of Sakamena samples and 40% of the Sakoa samples) of the material collected produced consistent palaeomagnetic results.

4. Comparison with African palaeomagnetic results

In our previous paper [1] we summarized the evidence relating to the various proposals that have been put forward concerning the drift history of Madagascar. We are now in a position to compare poles for three ages corresponding to the Sakoa, Sakamena and Isalo Groups with their equivalents from the African mainland. These can be compared according to the three possibilities usually considered tenable for Madagascar's past position adjacent to Africa.

(1) A position adjacent to Somalia, Kenya and Tanzania, originally strongly advocated by Du Toit [18] and favoured by Smith and Hallam [19] in their geometrical reconstruction. The position is obtained by rotating Madagascar about an Euler pole at 9°S, 313°E by an angle of 15°.

(2) A position adjacent to Mozambique occupying the coastal embayment as proposed by Flores [20]. This position is obtained by rotation about an Euler pole at 48°S, 70°E by an angle of 14° [1].

(3) It has remained in its present position.

The Upper Carboniferous to Lower Permian Sakoa Group palaeomagnetic pole can be compared with the mean of five poles from mainly Lower Permian formations sampled in Morocco [21] and East Africa [22] listed in Table 3. The Upper Permian to Lower Triassic Sakamena Group can be compared with the only two Early Triassic poles available for Africa listed recently by McElhinny and Brock [23]. The mean is also listed in Table 3. Twelve results were available [23] for the Late Triassic to Middle Jurassic of Africa and two new results have since been published giving results from Liberia [24] and South West Africa [25]. The mean of these 14 poles is given in Table 3 and may be compared with our previous result from the Isalo Group [1]. The angular differences between the Madagascar and African poles according to each of the three hypotheses outlined above are listed in Table 3. For all three sets of pole positions the minimum separation is obtained according to the Smith and Hallam [19] reconstruction (Table 3, Fig. 7).

Palaeomagnetic tests of the hypotheses relating to the palaeoposition of Madagascar have been made difficult because of the small angular differences involved in the comparison of palaeomagnetic poles between Africa and Madagascar. In our first result from the Isalo Group we believed the comparison to be statistically significant and in favour of the Smith and Hallam reconstruction. Since the palaeomagnetic poles from three different ages each favour the Smith and Hallam reconstruction we regard the palaeomagnetic compar-

TABLE 3

Angular differences between Madagascar poles and their nearest corresponding African poles according to three reconstructions

African pole (age)		Madagascar pole (age)		Angular differences (degrees)		
				present position	Smith-Hallam	Flores
37.9S, 61.9E	(Pl)	47.9S, 84.1E	(Cu-Pl)	19.0	9.8	17.6
67S, 88E	(Trl)	64.9S, 113.9E	(Pu-Trl)	10.7	10.2	15.4
67.6S, 73.8E	(Tru-Jm)	74.2S, 87.1E	(Tru-Jm)	10.0	5.3	13.7
Mean differences				13.2	8.4	15.6

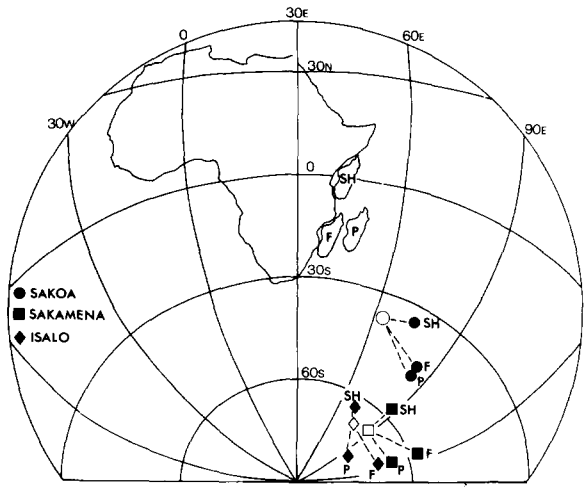


Fig. 7. Palaeomagnetic poles for the Sakoa, Sakamena and Isalo Groups (solid symbols) according to three reconstructions of the palaeoposition of Madagascar: P = present position; SH = Smith and Hallam reconstruction [19]; F = Flores reconstruction [20]. The corresponding African mean poles (Table 3) are shown as open symbols. Closest agreement is with the Smith and Hallam reconstruction (Table 3).

ison as now being conclusively in its favour. In our previous paper we summarized much of the geological evidence that also favoured such a reconstruction. Recently Wild [26] has examined the distributions of about 80 species of flowering plants common to Madagascar and the mainland of Africa in relation to the same three hypotheses that we have tested palaeomagnetically. He concludes that the distribution pattern as a whole also favours the more northerly position as on the Smith and Hallam reconstruction.

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